

Sahil Kumar

+91-8307539844

sahilkumar832832@gmail.com

Linkedin

Github

Portfolio

Professional Summary

Backend-focused Software Engineering Intern experienced in building scalable APIs and fault-tolerant systems. Strong foundation in data structures and distributed workflows, with hands-on experience designing asynchronous pipelines and AI-powered retrieval systems using RAG and vector search.

Technical Skills

Programming Languages: C++, Java, Python, JavaScript, TypeScript, SQL

Backend: Node.js, Express.js, RESTful API, Authentication & Authorization (JWT), MongoDB, PostgreSQL

Frontend: React.js, Redux, HTML, CSS, TailwindCSS

AI / Generative AI: LLM Integration, Prompt Engineering, RAG Pipelines, Embeddings, Vector Databases (Pinecone, FAISS), Semantic Search

Systems & Tools: Git, Github, Docker, Linux

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, DBMS

Experience

Software Development Intern — Rawyal

Node.js, Express.js, Python, JavaScript, RAG, Pinecone, ImageMagick, LibreOffice

- Architected a scalable **multi-format document processing pipeline** supporting Office, PDF, HTML, and image transformations, reducing external API dependency and improving processing speed by **35%**.
- Designed and deployed a **scalable asynchronous job-processing architecture** for concurrent document conversions, reducing peak request blocking time by **40%** and improving system throughput.
- Built a **Retrieval-Augmented Generation (RAG) chatbot** using **LLM embeddings and Pinecone vector search**, enabling contextual query answering across uploaded documents and internal knowledge bases.
- Developed optimized **Python-based extraction modules** for structured PDF-to-DOCX and Excel conversion, reducing manual correction effort by **30%**.
- Implemented **secure file validation, retry mechanisms, and failure isolation** to prevent cascading errors during malformed inputs and high-load processing.

Projects

LifeFlow – Real-Time Blood Donation Platform

GitHub | Live

Node.js, Express.js, MongoDB 2dsphere Indexing, Redis, Twilio API

- Designed and optimized a geospatial donor-matching system using **MongoDB 2dsphere indexing**, improving emergency search query performance by **50%**.
- Implemented **Redis-based caching** for frequently queried donor search results, reducing database load and improving response latency.
- Architected secure **role-based REST APIs with JWT authentication** and applied rate-limiting middleware to prevent abuse and ensure API stability under high traffic.
- Engineered an **asynchronous SMS notification pipeline** with retry logic and failure isolation to improve delivery reliability during network disruptions.

FarmFresh – AgriCommerce Platform

GitHub | Live

Node.js, Express.js, MongoDB, React.js

- Designed and implemented full-stack buyer-seller workflows with protected API routes and role-based access control.
- Built persistent authentication mechanisms and optimized state management to reduce session-related inconsistencies by **30%**.
- Developed an AI-assisted crop advisory module using structured prompt pipelines and response parsing for context-aware recommendations.

Education

Indian Institute of Information Technology, Una

2023 – Present

B.Tech in Electronics and Communication Engineering

CGPA: 7.04

Achievements

- Rated 3-Star (Max Rating: 1628) on CodeChef.
- Solved 300+ Data Structures and Algorithms problems across arrays, graphs, dynamic programming, and trees.
- Secured Global Rank 2200 in LeetCode Weekly Contest among 20,000+ participants.

Leadership & Initiative

Event Coordinator — EPMOC Club, IIIT Una

Oct 2025 – Present

- Led execution of technical and cultural events for 100+ participants, managing logistics and team coordination.